

The Tesora Harness

The framework beneath the Workbench

2026

Confidential

Tesora is an AI-native actuarial workbench for specialty P&C pricing and reserving. The **Workbench** is the surface an actuary works in, a chat next to Excel. The **Harness** is the framework beneath it that reads the data, runs the analysis, and makes the output something a chief actuary can sign. This note describes how the Harness is built, how it is kept honest, and how it handles your data.

01 One system, four narrow jobs

Tesora does not train one large model to do everything. The Harness coordinates four specialized agents, each with a narrow job, over the frontier models best suited to that job. Splitting the work this way is what makes the output checkable: a reviewer can see which agent produced which number and trace it back, rather than trusting a single black box that read the whole packet and returned an answer.

Every agent runs against **your data, in your environment**. Nothing about the design assumes a shared model that has memorized other carriers' books.



Every step writes to **one append-only audit trail**, and the actuary signs off before anything ships.

02 No single agent gets the last word

Ingestion Agent **READS**

Takes loss runs, statements of values, dec pages, and rate manuals in whatever format they arrive, including scans and inconsistent carrier templates, and normalizes them underneath the workflow you already run. It flags anomalies before they reach a model rather than after they reach a filing.

Insight Agent **MODELS**

Fits frequency and severity, trends, and on-levels, and assembles loss development. It hands back a write-up the reviewer can actually check, with the selections left to the actuary. It does the assembly, not the judgment.

Rating Agent **DEPLOYS**

Turns the actuary's selections into a versioned, tested rater, exposed as an API or a GUI, and generates the state exhibits that go with a filing. Existing rating logic is preserved exactly; changes ship with test cases attached.

Audit Agent **PROVES**

Traces every value to a page, a row, and a cell, and writes down anything it skipped, so a capped limit or a data gap lands in the memo where a reviewer will see it. The audit log is append-only.

03 Honesty is enforced, not requested

Three properties the Harness enforces in code, so they hold on every run whatever the prompt says.

CITATIONS OR IT DID NOT HAPPEN

Every value carries a cell-level citation to the source document. A number with no source is treated as a failure, not a result.

SELF-TESTING BEFORE DEPLOY

Models and raters ship with test cases and are re-run against them before anything goes live. The system reports what it could not verify.

THE ACTUARY OWNS THE DECISION

Agents assemble and check. Selections, sign-off, and the filing stay with a credentialed actuary at every step.

CALIBRATED ON REAL WORK

The Harness is trained on the CAS syllabus and the ASOPs a reviewer is held to, then tuned against live pricing, rater filings, and rebuilds reviewed by working actuaries.

04 Your data never leaves your boundary

Tesora runs in a **single-tenant VPC** scoped to your workflow. Production data does not leave that boundary, and Tesora uses AI providers configured for zero data retention, so your submissions and loss data are never used to train a shared model. Where documents are used to tune the system for your book, they are de-identified first, preserving structure while removing anything sensitive.

Single-tenant VPC

Zero data retention providers

SOC 2 ready

SOX controls

ASOP 23 / 41 / 56

NAIC Model Bulletin 2023-25

05 Signed the way a regulator expects

Rate changes get controlled, signed, and audited the way a regulator expects. Tesora enforces **author, reviewer, and approver as three separate people**, keeps an append-only record of every change, and ties each output back to the inputs it was built on. The result is work that moves at software speed and still survives an examination.

The short version: the Harness is designed so that a fast answer and a **defensible answer are the same answer**. Speed does not come at the cost of the audit trail, because the audit trail is how the system works, not a report it writes at the end.

NEXT STEP

See it run on your own book, in a 60-day build inside your environment.
Sign or refund.

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